Name Of Course	Bachelor Of Computer Application (BCA)
Semester	4TH
Name Of Subject	RDBMS
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Marks	75 Marks
Lectures	50 Lectures

Unit I

1. Introduction and Basic Concepts

1.1 Introduction to RDBMS

1.2 Characteristics of RDBMS

1.3 Applications and Advantages of RDBMS

1.4 Data models – Entity Relationship (ER), Mapping ER Model to Relational Mode, Network.

1.5 Relational and Object Oriented Data Models

1.1 Introduction to RDBMS:

An **RDBMS (Relational Database Management System)** is software used to manage and organize data in a relational database. It stores data in tables consisting of rows and columns, making it easy to access, retrieve, and manipulate using SQL (Structured Query Language).

The core features of RDBMS include:

1. Data Organization:

Data is stored in tables, allowing relationships between different data entities to be defined.

2. ACID Properties:

Ensures reliability and consistency of transactions (Atomicity, Consistency, Isolation, Durability).

3. Data Integrity:

Enforces rules like primary and foreign keys to maintain data accuracy.

4. Scalability:

Supports large datasets and multiple users simultaneously.

5. Popular RDBMS Examples:

MySQL, PostgreSQL, Oracle Database, and Microsoft SQL Server.

RDBMS is widely used for structured data in applications like banking, e-commerce, and healthcare systems.

1.2 Characteristics of RDBMS:

The **Characteristics of RDBMS (Relational Database Management System)** are as follows:

1. Data Storage in Tables:

Data is organized into rows and columns, with each row representing a record and each column representing an attribute.

2. Relationships Between Tables:

RDBMS uses keys (Primary Key and Foreign Key) to establish and enforce relationships between tables.

3. Use of SQL:

SQL (Structured Query Language) is used to query, insert, update, and delete data efficiently.

4. Data Integrity and Accuracy:

Ensures data consistency using constraints like primary keys, foreign keys, and unique constraints.

5. ACID Compliance:

Guarantees reliability during transactions by ensuring Atomicity, Consistency, Isolation, and Durability.

These characteristics make RDBMS a robust choice for managing structured data in various applications.

1.3 Applications and Advantages of RDBMS:

Applications of RDBMS:

1. Banking Systems:

Used to store customer accounts, transaction details, and manage funds securely.

2. E-Commerce:

Manages product catalogs, customer orders, and payment details.

3. Healthcare:

Stores patient records, appointments, and medical histories.

4. Education:

Tracks student data, courses, grades, and faculty information.

5. Government Systems:

Manages citizen records, tax details, and public services efficiently.

Advantages of RDBMS:

1. Data Consistency and Integrity:

Enforces rules to maintain accurate and consistent data using constraints.

2. Ease of Access:

SQL allows easy and flexible querying of data.

3. Data Security:

Provides user authentication and access control to protect sensitive data.

4. Supports Relationships:

Handles complex data relationships through keys and joins.

5. Scalability:

Accommodates growing datasets and multiple concurrent users effectively.

These features make RDBMS essential for modern applications requiring structured and reliable data management.

1.4 Data models – Entity Relationship (ER), Mapping ER Model to Relational Mode, Network:

Data Models in Database Management:

1. Entity-Relationship (ER) Model:

Represents data using **entities** (objects, such as a student or product) and **relationships** (associations between entities, such as enrollment or purchase).

Visualized through an ER Diagram with components like entities (rectangles), relationships (diamonds), and attributes (ovals).

Helps in designing a conceptual schema of the database.

2. Mapping ER Model to Relational Model:

Entities:

Each entity is converted into a table, where attributes become columns.

Relationships:

For **1:1 relationships**, a foreign key is added to one of the related entity tables.

For **1:N relationships**, the primary key of the "one" side becomes a foreign key in the "many" side.

For **M:N relationships**, a separate table is created containing foreign keys from both entities.

Primary Keys:

Ensures unique identification of records in relational tables.

3. Network Data Model:

Represents data as a graph with **nodes** (data entities) and **edges** (relationships).

Supports many-to-many relationships and is more flexible than the hierarchical model.

Uses **pointers** to link records, making data retrieval faster in complex relationships.

Example:

The CODASYL DBTG model.

Summary:

The **ER model** is used for conceptual design, while mapping transforms it into the relational model for implementation.

The **Network model** provides an alternative for representing complex relationships using pointers.

1.5 Relational and Object Oriented Data Models:

Relational Data Model:

The **Relational Data Model** organizes data into **tables (relations)**, consisting of rows (tuples) and columns (attributes).

Key Features:

1. Tabular Structure:

Data is represented in structured tables with clearly defined relationships.

2. Primary and Foreign Keys:

Used to ensure uniqueness and establish relationships between tables.

3. Data Integrity:

Constraints like primary keys, foreign keys, and unique constraints maintain consistency.

4. SQL Support:

SQL is used for data manipulation and querying.

5. ACID Properties:

Ensures reliable and consistent transactions.

Use Cases:

Banking, e-commerce, and other applications where structured data and strict relationships are needed.

Object-Oriented Data Model:

The **Object-Oriented Data Model (OODM)** combines database concepts with object-oriented programming principles.

Key Features:

1. Objects and Classes:

Data is represented as objects, encapsulating both data (attributes) and behavior (methods).

2. Inheritance:

Supports class hierarchies, allowing child classes to inherit properties from parent classes.

3. Encapsulation:

Data and methods are bundled together, improving modularity.

4. Relationships:

Supports complex relationships like aggregation and association.

5. Polymorphism:

Allows the same operation to work differently for different objects.

Use Cases:

CAD systems, multimedia applications, and software that require handling complex, unstructured data.

Comparison:

The **Relational model** is ideal for structured data with clear relationships, while the **Object-Oriented model** excels in managing complex, real-world entities and behaviors.